Network Protocols Reference

OSI Model Layers

The **OSI (Open Systems Interconnection)** model organizes network communication into seven layers, each responsible for a specific aspect of data transmission [Zimmermann, 1980].

Layer	Name	Protocol Examples	Data Unit
7	Application	HTTP, FTP, SMTP, DNS	Message/Data
6	Presentation	TLS/SSL, MIME, XDR	Message
5	Session	NetBIOS, RPC, SIP	Message
4	Transport	TCP, UDP, QUIC	Segment
3	Network	IP, ICMP, BGP, OSPF	Packet
2	Data Link	Ethernet, Wi-Fi (802.11)	Frame
1	Physical	Ethernet cable, Fiber, DSL	Bit

Transport Layer Protocols

TCP vs UDP vs QUIC

Feature	TCP	UDP	QUIC
Connection type	Connection-oriented	Connectionless	Connection-oriented
Reliability	Guaranteed delivery	Best-effort	Guaranteed delivery
Ordering	In-order delivery	No ordering	In-order (per stream)
Error correction	Yes	No (optional)	Yes
Congestion control	Yes	No	Yes
Head-of-line blocking	Yes	No	No (multiplexed)
Handshake latency	1.5 RTT	0 RTT	0–1 RTT
Typical use	HTTP, email, files	DNS, video	HTTP/3, mobile apps

> ***QUIC*** *was designed at Google to eliminate TCP's head-of-line blocking problem in multiplexed HTTP connections, reducing connection setup latency on lossy mobile networks [Langley et al., 2017].*

Application Layer Protocols

HTTP Version History

- **HTTP/1.0** (1996)
  - One request per TCP connection; connection closed after each response
  - No persistent connections by default

- No header compression in specification

- **HTTP/1.1** (1997)
  - Persistent connections (keep-alive) reduce connection overhead
  - Pipelining allows multiple requests without waiting for responses
  - Chunked transfer encoding for streaming responses
  - Required `Host` header enables virtual hosting on shared servers
- **HTTP/2** (2015)
  - Binary framing layer replaces text-based protocol
  - Multiplexing: multiple requests share a single TCP connection
  - Header compression using the HPACK algorithm
  - Server push allows proactive resource delivery to clients
  - Still suffers from TCP's head-of-line blocking at the transport layer
- **HTTP/3** (2022)
  - Runs over QUIC instead of TCP, eliminating head-of-line blocking
  - Faster connection establishment with 0-RTT resumption for known servers
  - Improved performance on high-latency or lossy mobile networks
  - Independently retransmits lost packets per stream without blocking others

DNS Resolution Process

DNS (Domain Name System) translates human-readable hostnames to IP addresses through a hierarchical delegation chain:

1. **Client Query**
   - Browser checks its local in-memory cache
   - OS resolver checks `/etc/hosts`, then the system DNS cache
   - If not resolved, forwards query to the configured recursive resolver
2. **Recursive Resolver**
   - ISP or public resolver (e.g., 8.8.8.8 or 1.1.1.1) receives the query
   - Checks its own cache; on a miss, begins iterative resolution
3. **Root Name Servers**
   - 13 root server clusters worldwide (a.root-servers.net through m.root-servers.net)
   - Returns NS records for the appropriate top-level domain (e.g., `.com`, `.org`)
4. **TLD Name Servers**
   - Operated by registries (e.g., Verisign for `.com`)
   - Returns NS records pointing to the domain's authoritative nameserver
5. **Authoritative Name Server**
   - Holds the zone file for the queried domain
   - Returns the final A or AAAA record (IPv4 or IPv6 address)
   - The TTL field controls how long resolvers may cache the answer

Security Protocols

Protocol	OSI Layer	Purpose	Successor To
TLS 1.3	4/5	Transport encryption	TLS 1.2, SSL
SSH v2	7	Secure remote shell	Telnet, rlogin

IPsec	3	IP-level VPN encryption	—
DNSSEC	7	DNS record signing	Plain DNS
HTTPS	7	HTTP over TLS	HTTP

References

- [Zimmermann, 1980] H. Zimmermann, "OSI Reference Model — The ISO Model of Architecture for Open Systems Interconnection," *IEEE Trans. Communications*, 1980.
- [Langley et al., 2017] A. Langley et al., "The QUIC Transport Protocol: Design and Internet-Scale Deployment," *SIGCOMM*, 2017.